IBM Z DATATHON 2026 · THEME 2: AI SECURED

Los Antimarrulla

Privacy-preserving screening

AI estimates diabetes risk from encrypted inputs. Designed for IBM LinuxONE; local results available today.

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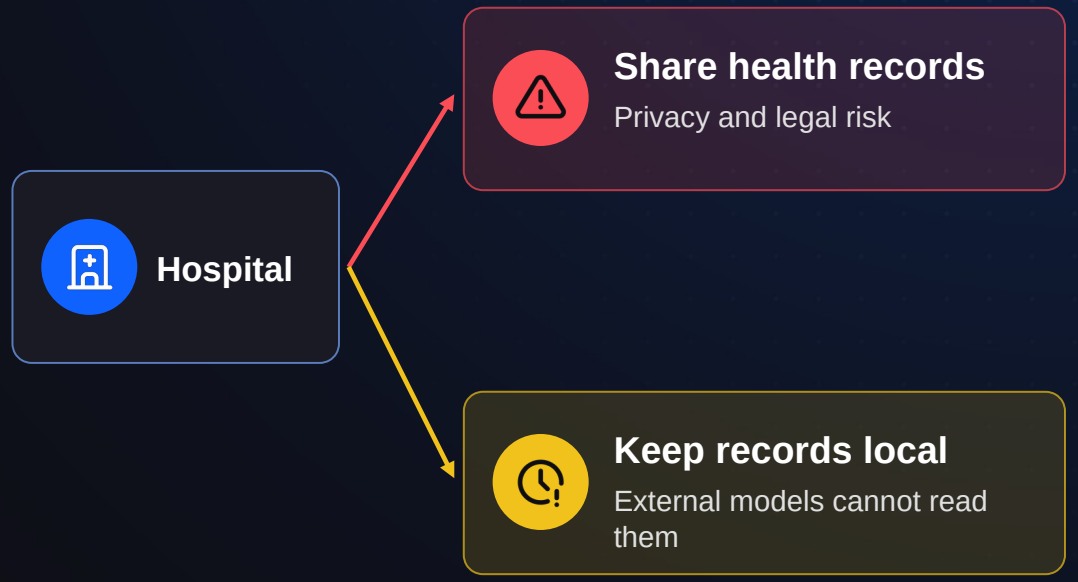
01 · THE PROBLEM

Diabetes is widespread; health records are sensitive

18.3% of adults in Mexico live with diabetes

1 in 3 people with diabetes are unaware of it

112,577 deaths in 2024: Mexico's second leading cause of death

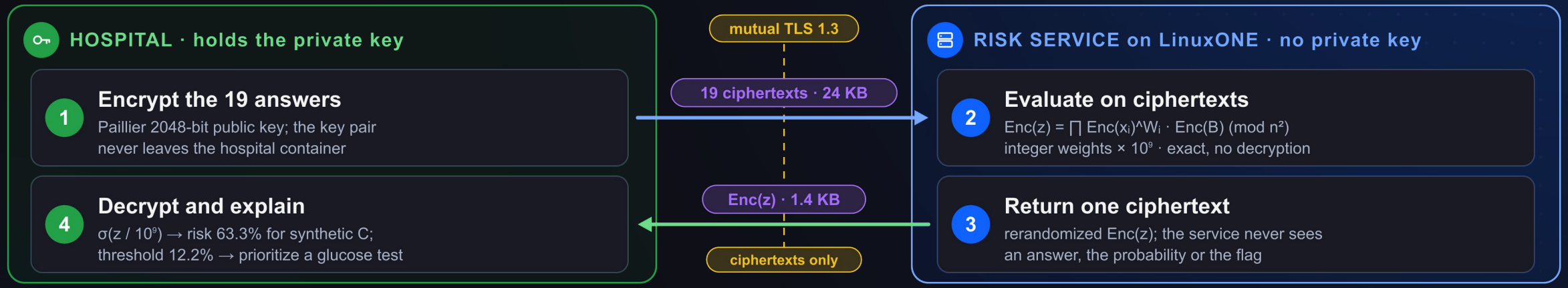


Can AI calculate risk without reading the patient's inputs?

Sources: ENSANUT 2022 (Basto-Abreu et al., Salud Pública de México, 2023); 2024 registered deaths and SSA hospital discharge microdata, DGIS/INEGI.

02 · THE SOLUTION

Calculate risk from encrypted inputs



Ciphertexts only

The risk service receives no readable age, BMI, blood pressure or other feature

Same ranking

Local sample: encrypted AUC equals plaintext AUC

Screening support

Prioritize confirmatory glucose testing; this is not a diagnosis

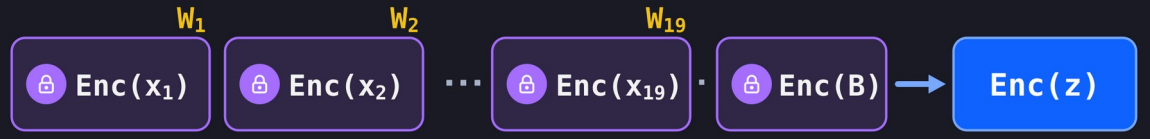
AI sees the risk. Nobody sees the patient.

Scope: the risk service receives ciphertexts only. The planned LinuxONE demo uses two containers on one VM with a trusted administrator and synthetic patients. VM execution is still pending.

Calculate a score without decrypting inputs

THE PAILLIER PROPERTY

$\mathrm{Enc}(a)\cdot\mathrm{Enc}(b) = \mathrm{Enc}(a+b)$ $\qquad$ $\mathrm{Enc}(a)^k = \mathrm{Enc}(k\cdot a)$



Hospital decrypts $z = W\cdot x + B \rightarrow \text{risk} = \sigma(z / 10^9)$

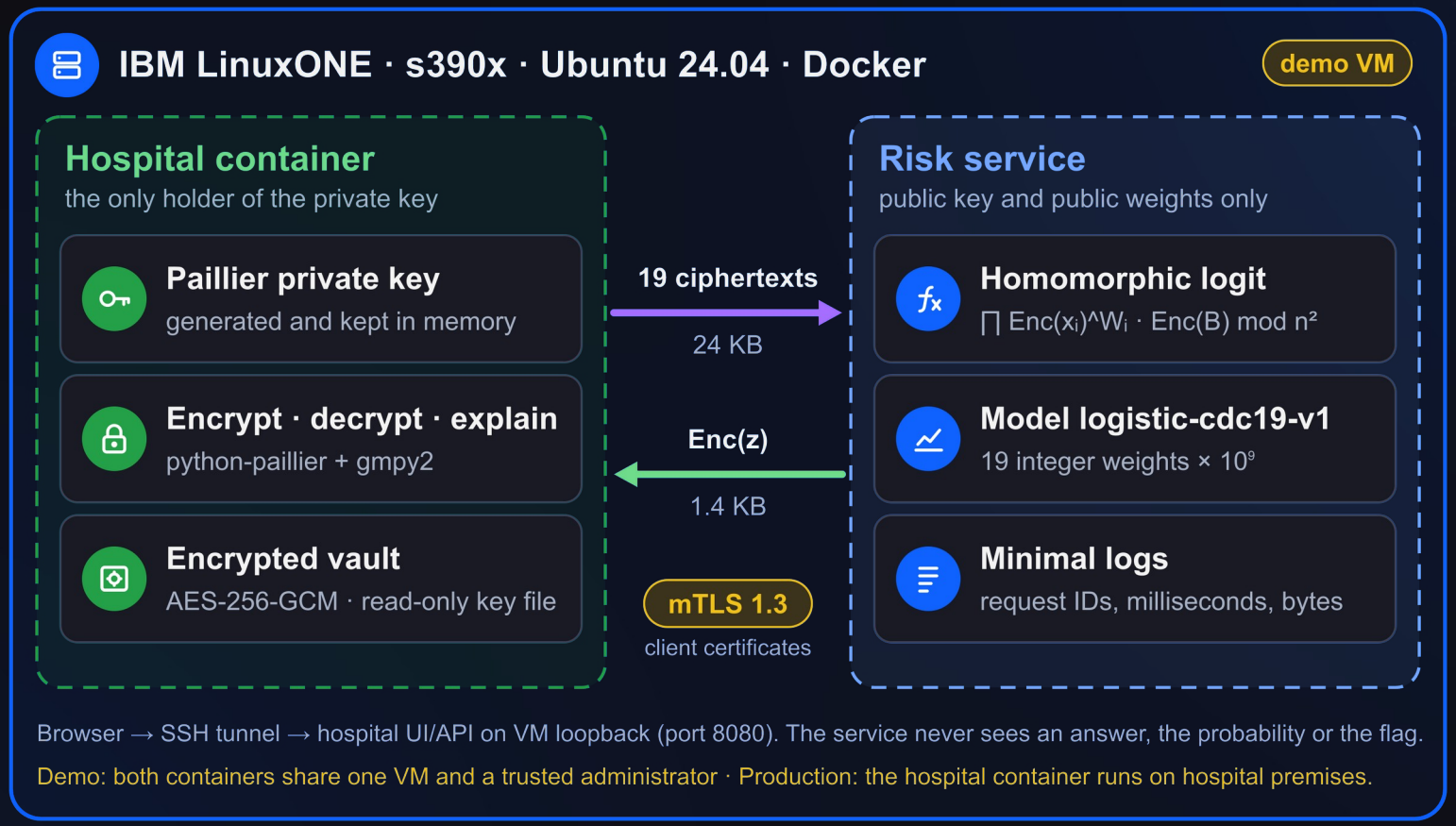
Exact: decrypted z matched the fixed-point reference on 2,000/2,000 CDC records.

In transit	Mutual TLS 1.3 between hospital and service
At rest	AES-256-GCM in the hospital vault
During use	2048-bit Paillier: calculation without decryption

Who can see what?

Component	Patient inputs	Risk result
Hospital (private-key holder)	Plaintext: its own patients	Plaintext
Risk service (no private key)	**Ciphertexts only**	**Ciphertext only**
Hospital vault records	**AES-256-GCM ciphertext**	**AES-256-GCM ciphertext**
Service logs	**No: IDs, ms and byte counts only**	**No**

Two containers; the private key stays at the hospital



Large-integer workload

Modular exponentiation over integers thousands of bits long for every patient

Isolation and encryption

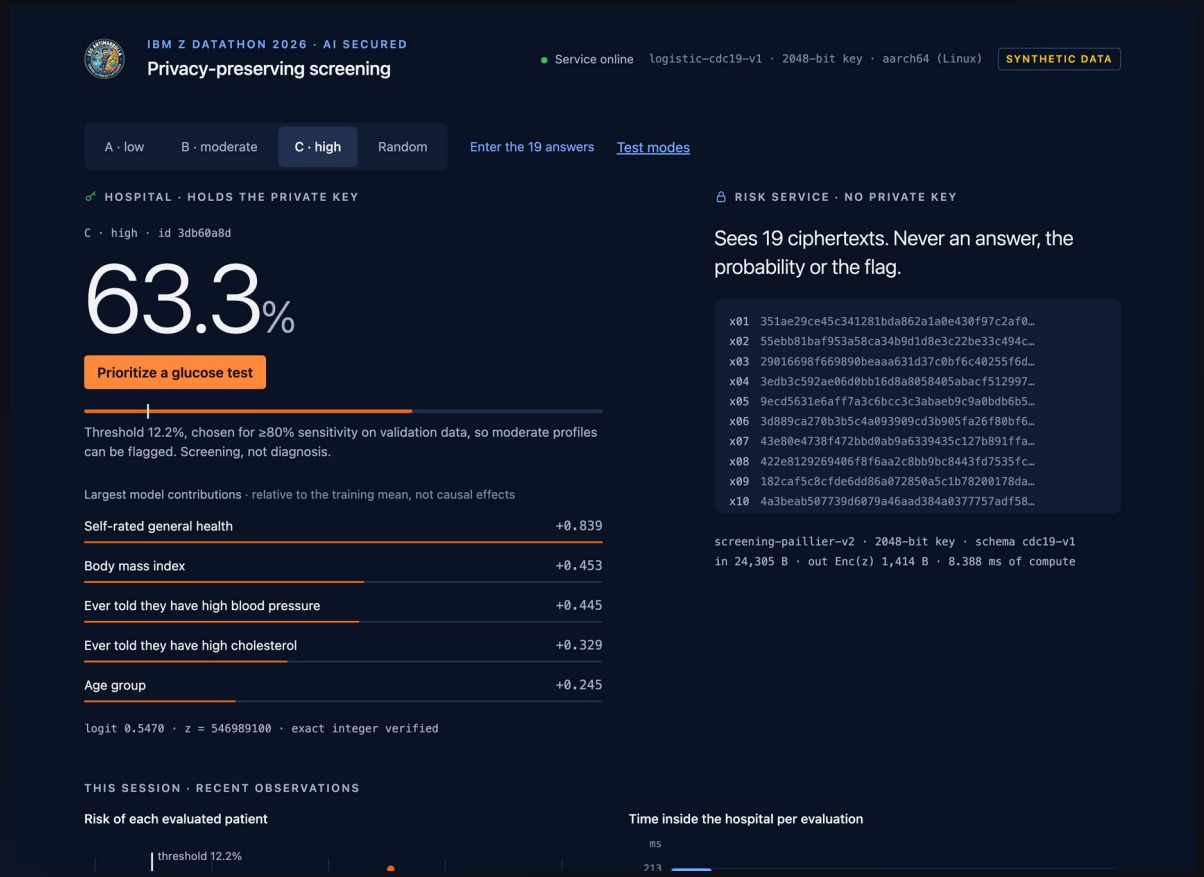
Separate containers and mutual TLS; effective CPACF use has not been measured

Measure the actual platform

Local results available; repeat the same benchmark on LinuxONE

Stack: Python · scikit-learn · python-paillier + gmpy2 · cryptography (TLS 1.3, AES-256-GCM) · Docker (Ubuntu 24.04, s390x)

The service sees ciphertexts; the hospital sees risk



Actual English UI with a synthetic patient · local containers · LinuxONE run pending

2,000 of 2,000
decrypted integers exactly match the fixed-point reference

ROC-AUC 0.807
equal across float, fixed point and encryption on the same 2,000 records

115 ms (p50)
Hospital API: encrypt, mTLS, evaluate, decrypt and vault write (200 synthetic requests)

24.3 KB → 1.4 KB
19 input ciphertexts → one encrypted score

LOCAL benchmark · model logistic-cdc19-v1 · commit be9b27e · macOS arm64 / Docker · 2048-bit key. API p95: 118 ms. LinuxONE run pending; repeat with make measure-api-docker.

Every screen shows what stays private

1 · What each side sees

Hospital answer vs. the ciphertext the risk service receives

ANSWER BY ANSWER

What the hospital encrypted and what the risk service received for each question, in protocol order.

#	Question	Hospital answer	Risk service receives
1	Ever told they have high blood pressure	Yes (1)	5b15a99f1fc3d67c0f2d1ea3…
2	Ever told they have high cholesterol	Yes (1)	2a2395b60283c313bc9fa0a8…
3	Cholesterol check in the past 5 years	Yes (1)	4892067dc61de3adfd238ea4…
4	Body mass index	36 kg/m²	146753b90f891abceb59849c…
5	Smoked at least 100 cigarettes in their life	Yes (1)	1d472757671844d1aedbe8ef…
6	Ever told they had a stroke	No (0)	3e603e6a38ba599635fb0b8d…

2 · One encrypted round trip

Encryption dominates: 19 × 2048-bit Paillier, then ~10 ms of service time

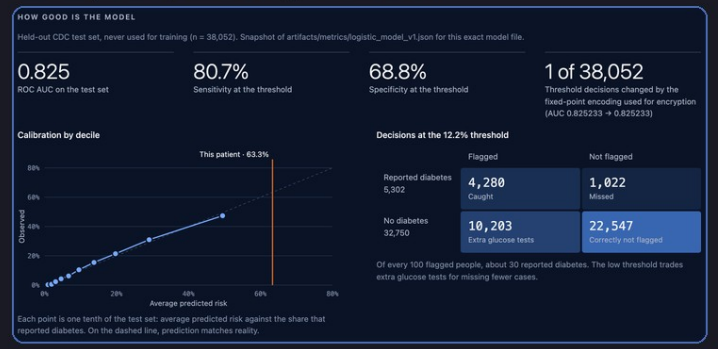
ONE ENCRYPTED ROUND TRIP

Step	Where	ms
Encrypt the 19 answers	Hospital	124.9
mTLS 1.3 + encrypted scoring	Network + risk service	10.2
Decrypt the score	Hospital	1.5
AES-GCM vault + fsync	Hospital	1.6
Total inside the hospital	**Excluding the browser**	138.2

Encryption takes the longest: each of the 19 answers is a separate 2048-bit Paillier encryption.

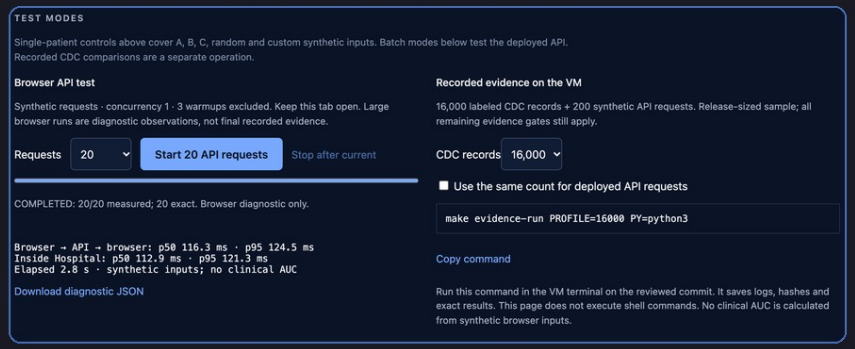
3 · Model quality in the app

38,052 held-out records; fixed point changes 1 decision



4 · Built-in test modes

20/20 exact browser API run; one command records VM evidence



Actual captures of the Hospital app · model logistic-cdc19-v1 · synthetic inputs · local containers; the same UI runs on the LinuxONE VM

Train on public data; evaluate encrypted patient inputs

253,680 public CDC BRFSS 2015 health survey records (UCI ID 891)

19 indicators: BMI, blood pressure, cholesterol, activity, age… (Education and Income removed)

13.9% self-reported prediabetes or diabetes: account for class imbalance

1. **Train on public records**
   Logistic regression; standardization folded into model weights
2. **Quantize the weights**
   Round weights × 10^9; all 19 inputs are already integers
3. **Encrypt patient inputs**
   Encrypt 19 indicators using the hospital's key
4. **Encrypted score → risk**
   The service evaluates w · x on ciphertexts; the hospital decrypts and applies sigmoid

TEST RESULTS (38,052 RECORDS)

Metrics

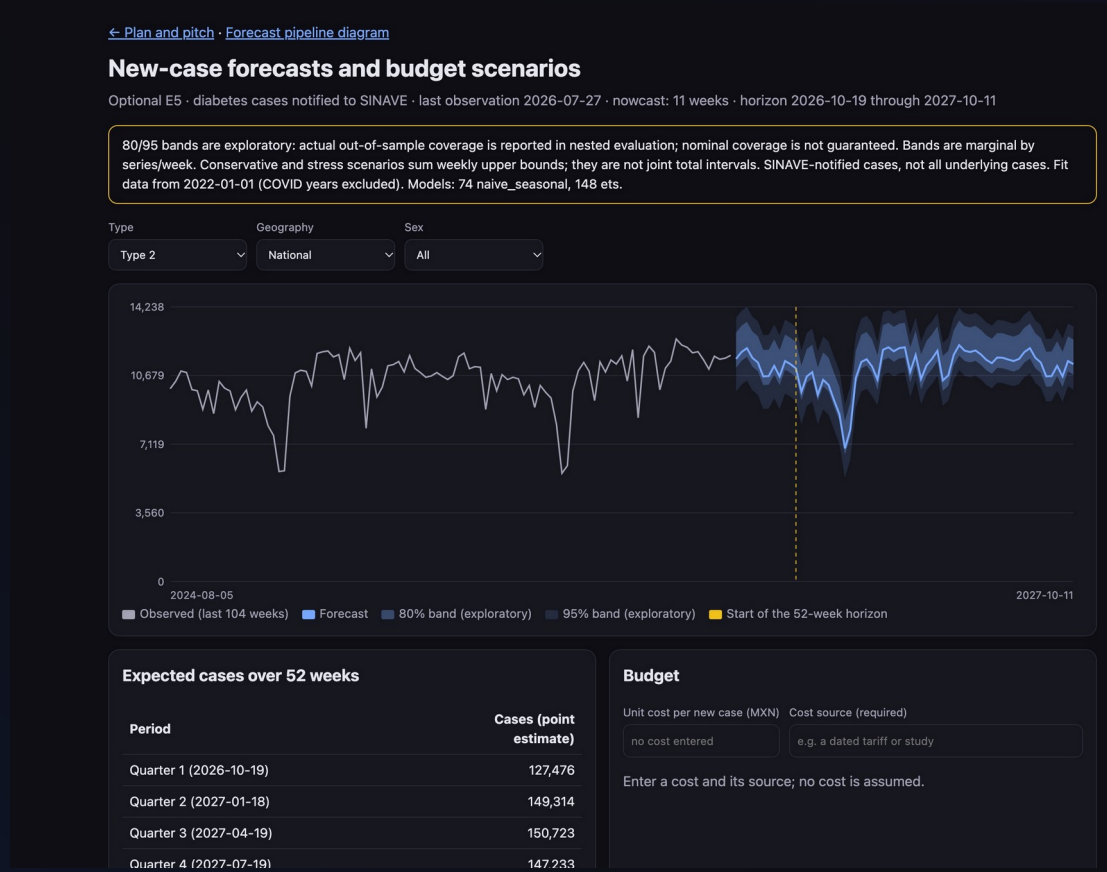
ROC-AUC 0.825 · AP 0.414 · sensitivity 0.807 · specificity 0.689 · PPV 0.296 · Brier 0.098. Threshold selected on validation.

Limitations

Label: self-reported U.S. diagnosis, not future diabetes. A separate reduced ten-feature ENSANUT model reaches AUC ≈0.77 and underestimates risk.

Source: UCI Machine Learning Repository, CDC Diabetes Health Indicators (ID 891).

Plan demand: weekly diabetes forecasts by state



Live dashboard · Type 2 · national · all sexes · antimarrulla-tamizaje.netlify.app/forecast

222 weekly series

T1/T2 × national, 4 regions and 32 states × men/women/all, reconciled top-down

SMAPE 13% national

nested out-of-sample test, weeks 12–63; regions 25%, states 60% (sparse type 1)

52 weeks ahead

Oct 19, 2026 – Oct 11, 2027 after an 11-week nowcast; ETS or seasonal naive per series

Budget = cases × cost

shown only with a sourced unit cost; no cost is assumed

EXPLORATORY · SINAVE-notified cases, not all cases · nominal 80% bands cover ≈64% in practice · public aggregates, no patient data · not yet run on LinuxONE.

09 · ROADMAP

From a verified demo to a hospital pilot

0	plaintext patient inputs reach the external risk service
4	test profiles ready for LinuxONE: 20 to 16,000 patients
222	weekly forecast series ready for capacity and budget planning

1. **Oct 2026 · Datathon MVP**
   Encrypted screening MVP; LinuxONE runs recorded with logs and CI checks
2. **Q1 2027 · Calibrate for Mexico**
   Recalibrate on Mexican data (ENSANUT) and benchmark s390x at scale
3. **Q2 2027 · Hospital pilot**
   Hospital container on site, risk service on LinuxONE; de-identified records
4. **Q4 2027 · Scale**
   Non-linear models with CKKS, Granite explanations at the hospital, forecasts on LinuxONE

WHAT IT TAKES

Partners

A public hospital or clinic network for the pilot, LinuxONE capacity and a sourced unit cost for budgets.

Gates before clinical use

Clinical validation in Mexico, legal and privacy review, an independent security audit and a stronger host trust model.

A plan, not a commitment: dates assume partner access and funding.

Screening support without sharing readable records

Earlier screening

Support earlier confirmatory testing for people who may be unaware of diabetes

Patient trust

The external risk service receives encrypted inputs

Privacy safeguards

A privacy-preserving design; clinical and legal review remain necessary

Further applications

Explore other linear screening models after separate validation

AI sees the risk. Nobody sees the patient.



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We challenge idleness and procrastination
github.com/UACJ-MIAAD/antimarrulla-ibmz-datathon-2026

Thank you

Screening support does not replace medical diagnosis.